

Basic Jogging and Axis Movement Experiment on Mitsubishi 6-Axis Robot

Aim

The aim is to become familiar with jog control using the teaching pendant, moving the robot in all six axes (J1-J6) and in different jog modes such as JOINT, XYZ (Cartesian), and TOOL coordinates.

Materials Required

- Mitsubishi 6-axis 8kg robot (e.g., RV-8CRL)
- Robot controller
- Teaching pendant with jog capability
- Safety barriers/fencing

Procedure

1. Preparation and Safety
 - Ensure the robot is powered ON, in TEACH mode, and the work area is clear.
 - Have all operators aware of the emergency stop locations.
 - Confirm the deadman switch and safety interlocks are working.
2. Jog Mode Selection
 - On the teaching pendant, press the [JOG] key to enter jog mode.
 - Select the desired jog mode: JOINT (axis-by-axis), XYZ, or TOOL using the pendant screen or dedicated buttons.
3. Axis Movement
 - Hold the deadman switch to enable servo power.
 - Select an axis (e.g., J1) on the pendant and use the [+] or [-] to move the respective joint. Observe movement.
 - Repeat for all axes (J1–J6).
 - Try XYZ mode to move along Cartesian axes, and TOOL mode for tool coordinate movement.

Image 1: Mitsubishi 6-Axis Industrial Robot Joint Diagram

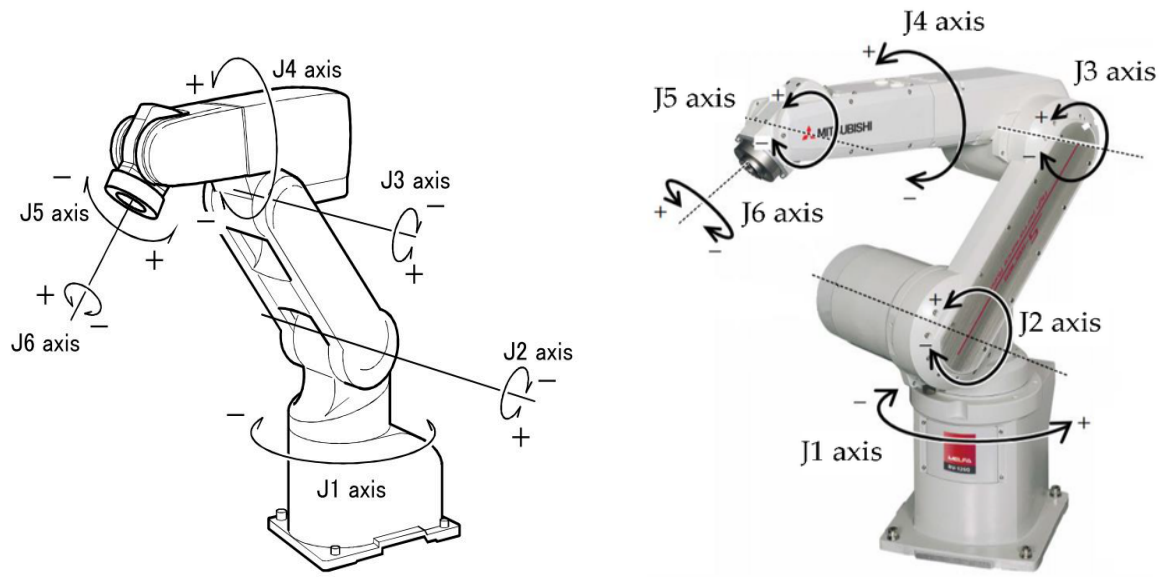


Image 2: Robot Teach Pendant



4. Speed Adjustment

- Adjust jog speed using the [OVRD] keys and verify speed display on the pendant.
- Operate at low speed for safety, gradually increasing if safe and necessary.

5. Emergency Stop Validation

- During movement, release the deadman switch to confirm the robot stops immediately.
- Press the emergency stop to verify full power-down response.

Result

This experiment covers fundamental manual movements, critical for teaching and troubleshooting. Ensure strict adherence to safety procedures throughout.